

LZ CABLE PLANNER

USER MANUAL

Plan, document and hand over broadcast and AV cabling: signal flow, rooms and floors, cable lengths, patch lists and the paperwork for the build day.

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1. Installation

Download the installer from the [latest release](#):

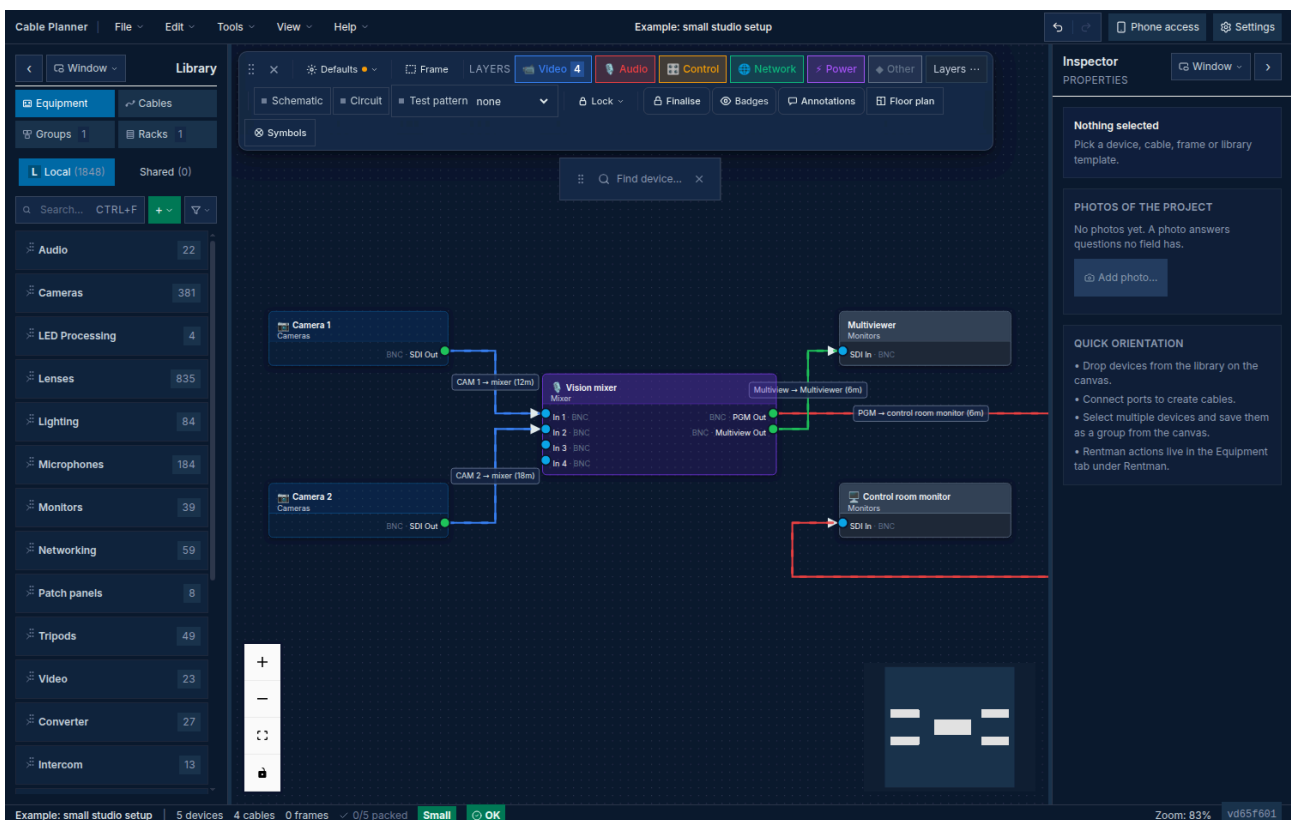
- **macOS:** `.dmg` for Apple Silicon and Intel
- **Windows:** `.exe`

The app runs fully offline. An internet connection is only needed for the device library, the cloud copy and live sessions across networks.

Without installing anything, the web edition runs at <https://larszu.github.io/cable-planner/> (see chapter 21 for what it cannot do).

2. First project

- **File** → **New** starts an empty plan. **Open**, **Save** and **Save As** work as usual; the recent projects list fills itself.
- A project is a single JSON file (`.cableplan`). It can be versioned with git and diffed like text.
- While you work, the plan is auto-saved into a recovery copy every few hundred milliseconds. If the project grows beyond what that store holds (about 5 MB), the status bar shows **"No recovery copy"** with the project size — then save to a file. The plan itself is unaffected.
- Undo and redo cover the last 100 steps.

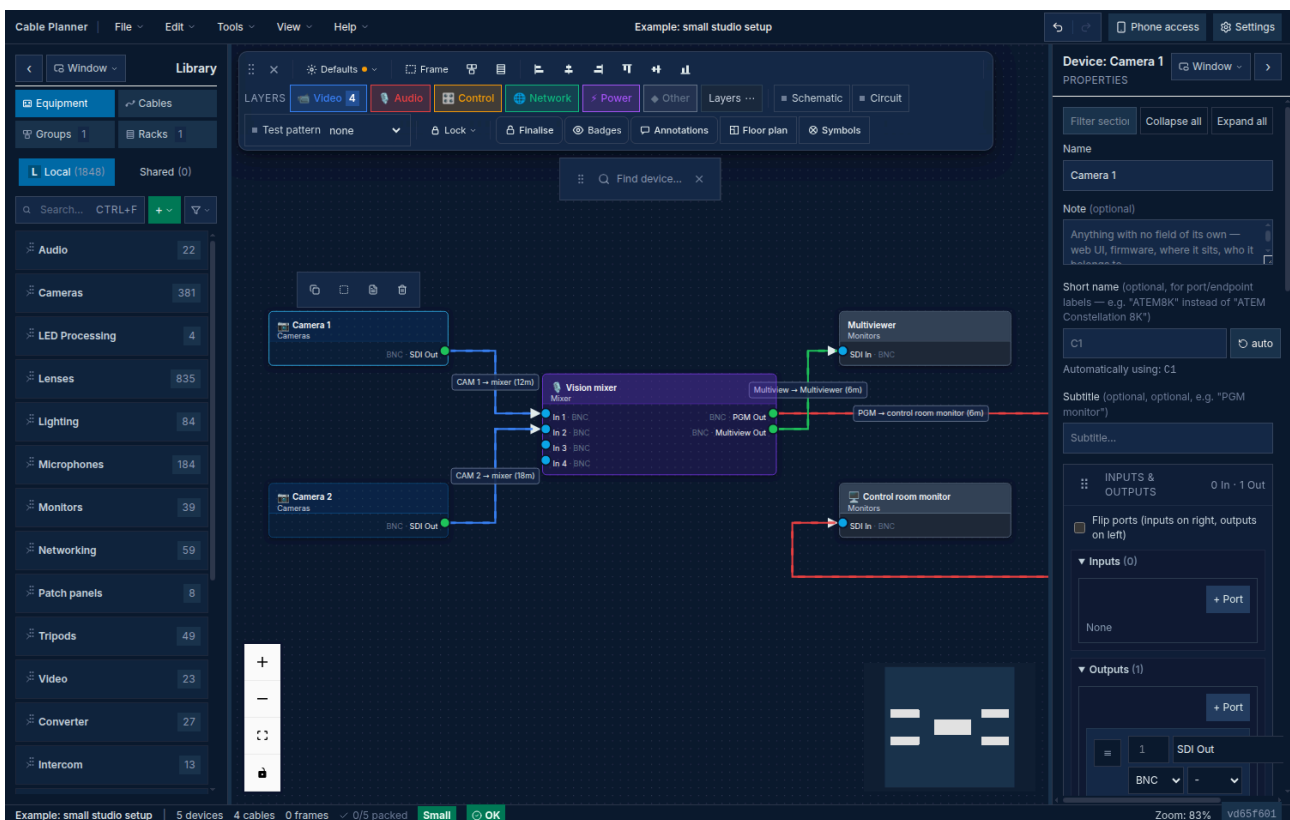


3. The canvas

- Drag devices from the **Library** onto the canvas. Zoom, pan and the minimap work with mouse, trackpad or touch.
- Drag from a port to a port to draw a cable. Bend points and segments follow the pointer for the whole drag.
- Right-click a device (long press on a touch screen) for its context menu.
- Right-click an empty spot → **New device here ...** creates a device for this project only.
- **Show signal path** (in a cable's properties) highlights the whole chain the cable belongs to — through plates, house runs, converters and routers — and lists every station with floor, room, device and port. **Esc** or the chip in the toolbar ends it.

4. Devices

Properties panel



The top of the panel is fixed: **name**, **note** (web UI, firmware, location, owner) and **inputs & outputs**. All other sections can be dragged into any order. The filter field finds a section by title; **Collapse all** / **Expand all** in one click. Open sections are remembered.

Creating your own device

- **Library → Create your own device** needs only a name. Ports, manufacturer and datasheet can follow later. Drop a cable end onto the body of such a device and it gets a matching port.
- **Place in project only** keeps the device out of the template library — for loan gear, a client's box or a placeholder.
- **Ports from a photo:** pick, drop or paste photos of the connector side (on a phone: *Take photo*). The AI provider chosen under *Settings → Integrations → AI* reads the connectors and the rating plate. You get an editable suggestion list; unsure rows are unticked, nothing is applied until you confirm. If the model exists in the built-in catalogue, that entry is offered instead.

Catalogue link

Devices created by hand, imported or loaded from older projects are linked to their catalogue model automatically when the name matches exactly one entry. Several matches are never guessed; the section *Catalogue & source* offers them. Catalogue entries carry a **manufacturer link** to the datasheet.

Entries whose sockets are not known are marked *ports unknown*. The plan check asks for the datasheet; the mark disappears as soon as you add real ports.

Network and streams

- **Network & access:** IP, mask, gateway, MAC and **VLAN**. The device card shows `10.0.0.5 /24 · VLAN 30`.
- **Streams:** what a device sends or receives (RTSP, SRT, RTMP, NDI, HLS, MJPEG, WebRTC, ST 2110, Dante, AES67 ...) with address, port, codec and format.
- **Credentials never enter the plan.** A user name, password or token typed into a stream address is removed when leaving the field. The desktop app keeps it in the computer's keychain.
- **Still image preview** under a device: from the camera's snapshot URL or — for an RTSP/RTMP/SRT/HLS/MJPEG stream — one frame via **ffmpeg** (install it separately: `brew install ffmpeg` or `ffmpeg.exe` on the PATH). Desktop app only, local network only, refreshed every 10 s. A preview starts only after you click **Start preview**.

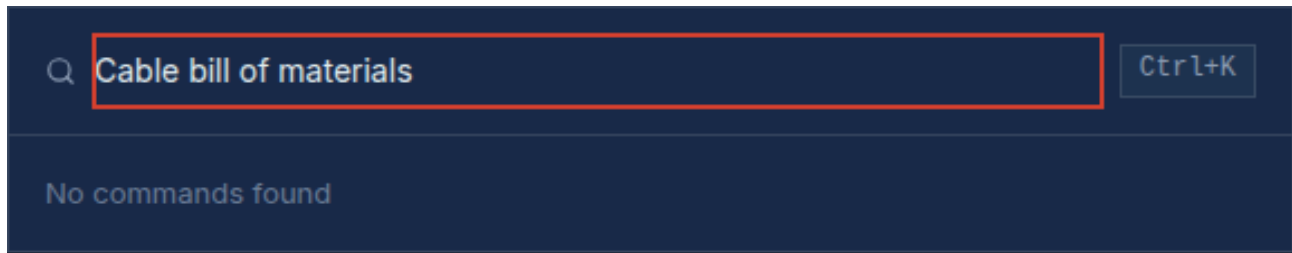
Master data

Settings → Master data holds your own connector types, signal standards and cable layers next to the built-in ones. Renaming carries the new name onto all ports, cables and templates of the open project.

5. Cables

- Every cable has type, length, colour, label and notes.
- The cable's properties, the pull list and the cable schedule show both ends as *floor · room · device · port*.
- **Stock lengths:** enter the drums you own under *Settings → Project → Available stock lengths*. The bill of materials then splits a 137 m run into 100 + 50 with one coupler — fewest couplers first, least

excess second — and names what the stock does not cover.



6. Rooms, floors and 3D

- A **frame** on the canvas is a room. It picks its floor from the project's floor list (*Floors* in the frame's properties, bottom to top, with the level in metres).
- **Rooms ▾** in the toolbar hides floors or single rooms. A cable into a hidden room remains as a stub that names its target; exports always stay complete.
- **3D** shows the building: every room at its floor height, the devices inside, the connections between rooms as one line per room pair or as single cables.
- Tick **Riser** on a frame to make it a shaft through all floors. Cables between floors then run via the nearest riser.

7. Floor plan and real cable lengths

Toolbar → **Floor plan**.

1. Load or drop an image (PNG, JPG, WebP, GIF, BMP, AVIF) — or import the venue from the MultiCam or Light Planner. PDF plans: export the page as PNG or JPG first.
2. Set the scale:
 - **Two points** on a plan drawn from above: click both ends of a known distance.
 - **Four corners** on a photo or an isometric drawing: click the corners of a floor area of known width and depth.
3. After calibration the plan locks and lies under everything.

Cable lengths then follow the drawn route, socket to socket, through every bend point. A length is marked **out of date** as soon as a device moves, a cable is re-routed or the plan is re-calibrated.

8. Symbols

Toolbar → **Symbols**: electrical, intrusion and fire alarm, voice alarm, IT, automation and AV, drawn after DIN EN 60617 and DIN 14034-6.

- A symbol is a label on the plan, not a device: no ports, no checks.
- Import your own as SVG, PNG, JPG or WebP; they travel inside the project file.

- With an AI key under *Settings* → *AI*, a symbol can be generated from a description.
 - The symbol list exports as CSV.
-

9. Adapters, gender changers, converters

- **Adapter** — changes the plug shape (BNC to RCA).
- **Gender changer** — changes pin or socket only.
- **Converter** — changes the signal (SDI to HDMI).

Adapters and gender changers can be inserted with one click on a selected cable; one undo takes the insertion back. Converters are **named**, not inserted: model, bandwidth and price are your decision. Inserted parts stay as an *open point* in the plan check until you have checked them.

10. Faceplates

Tools → *Faceplate editor...* — wall panels, stage boxes and rack plates.

- Place each connector in **millimetres**; the connectors are the device's own ports.
 - Enter the cutout diameter from the manufacturer's document — it is never guessed. Without it, holes are not checked against each other, and the report says so.
 - Checks: holes over the edge, overlapping holes (with the overlap in mm), connectors without a position.
 - Print the **label strip** (one per row) and the **drilling sheet** at 1:1.
 - A plate passes the signal through — socket *n* at the back to socket *n* at the front. Untick **Patch panel** for a stagebox with a converter inside. **On the plate** chooses which side is drilled.
-

11. Fibre breakouts and polarity

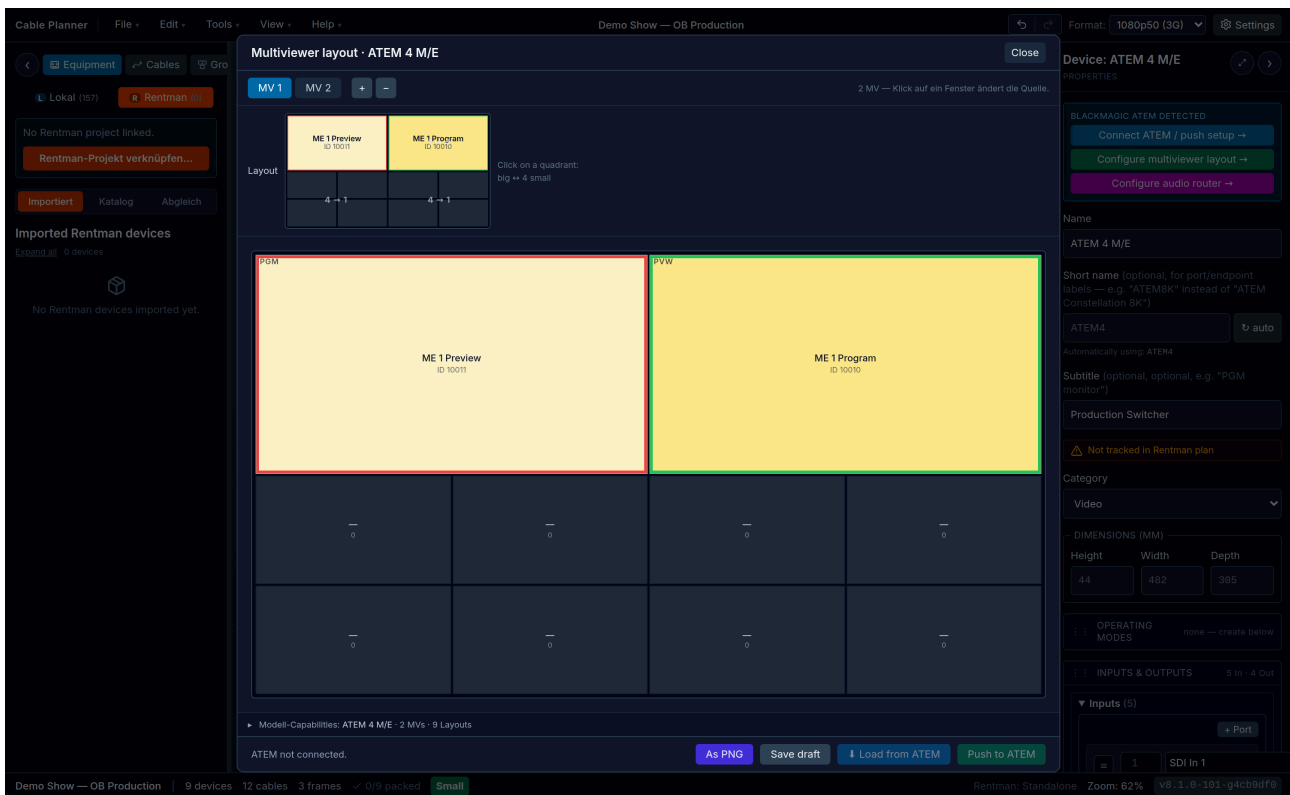
- A socket can carry a **breakout**: each fibre with position, **TX** / **RX** / *not stated* and its own tail connector.
 - Each cable end names the fibre it uses; patch and pull lists show it.
 - The plan check finds partly patched breakouts, two cables on one fibre and fibres the socket does not have.
 - **Polarity** is checked only against the method you choose (TIA-568 A, B or C). Without a method it is reported as *unchecked*.
-

12. LED walls

Tools → *LED wall...*

- Panel types with pixel pitch, resolution, size and — where stated — weight and power (average and peak).
- Enter the opening in millimetres: you get the grid and the leftover.
- Totals for panel count, resolution, size, weight and load; sending card ports needed against available.
- The wall is fed from a building outlet: its continuous load joins that outlet, its peak gets a finding of its own.
- **Pixel map** as PNG in the wall's native resolution, tiles numbered from the top left.

13. ATEM and Videohub



- **ATEM multiviewer** (*Tools → ATEM multiviewer layout...*): visual layout editor, program/preview and camera assignment, exportable configuration. For Television Studio, Constellation and M/E models.
- **Videohub** (*Tools → Videohub routing / labels...*): source → destination mapping, router configuration, routing overview and export.

The planner reads switchers and routers. It never sends switching commands from the AI interface.

14. Lists and the report editor

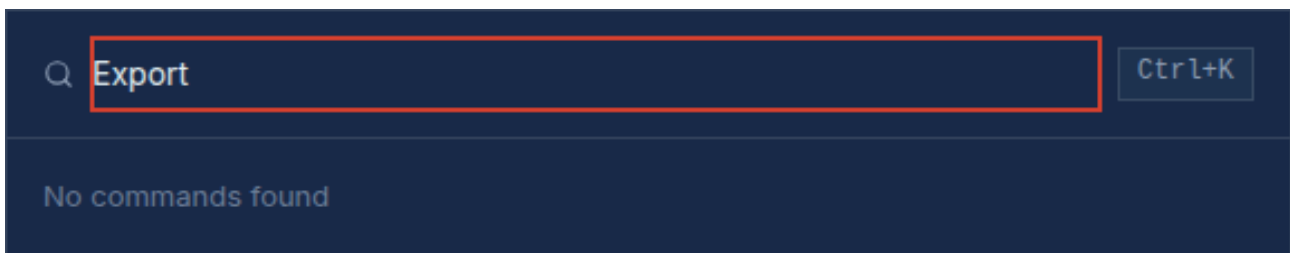
Every list — pull list, termination list, cable schedule, asset register, network sheet, spectrum plan, delivery, tally map, handover, signal paths, house run occupancy, crossings, defects, maintenance, configuration, attachments, faceplates — opens in one editor (*Tools → Report editor...*):

- show, hide and reorder **columns**
- **group**, **sort** by several columns, **filter** per column
- save **templates** for this project or for all projects

The preview is the export: screen, CSV and print come from the same result.

Patch list (*Tools* → *Patch list...*): one row per network interface, from the device port through every patch panel (with both port numbers) to the switch port, with IP, subnet, gateway, MAC and VLAN. Where a path does not reach a switch, the row says why.

15. Export and handover documents



- **File** → **Export & Print...**: PDF, PNG, JPEG, SVG with layer filter.
- **Per-device patch sheets** with location and the far end of each cable.

Patch list							
Search (device, port, type, colour, note ...)							All layers 4 / 4
No.	From device ^	Port ^	To device	Port	Type	Length (m)	Colour
	Camera 1	SDI Out	Vision mixer	In 1	BNC	12	
	Camera 2	SDI Out	Vision mixer	In 2	BNC	18	
	Vision mixer	Multiview Out	Multiviewer	SDI In	BNC	6	
	Vision mixer	PGM Out	Control room monitor	SDI In	BNC	6	
Each cable as its own row, sorted for patching order on set. more				Export CSV	Export XLSX	Labels + QR (PDF)	Generic (CSV) Label CSV

Tools → **Fixed install: docs & handover...**:

Document	Format
Pull list, termination list, cable schedule, BOM with reserve, asset register	CSV / PDF
Signal paths, house run occupancy	CSV
Route plan per floor	HTML, A4 landscape
Crossings (fire protection)	CSV
Acceptance record with defects, open items and signature block	HTML
Maintenance schedule	CSV
Configuration settings per interface	CSV
Device cards, operator overview, camera positions	HTML
Attachment index with checksums	CSV
QR labels for cables and devices	print

Device datasheet (device properties → *Print / documentation*): one A4 page per device with photo and the properties you tick. For several devices: *Export* → *Patch sheets* → *Datasheets*.

Attachments (desktop app): test reports, manuals and configuration backups, attached to a cable, a device or the installation. Files are copied into the folder **Anhaenge** next to the project.

Every exported CSV ends with a column glossary.

16. On site: phone, survey, labels

- **Phone access**: scan the QR code on the build day — no install, no account. Choose **Read only** or **Contribute**. Contributing phones tick off work, add pulled cables, file change requests and send photos; everything comes back into the plan. Requires the desktop app on the same network.
- **Photos** point at a device, a cable or the project and travel inside the plan file.
- **Survey** (*Tools* → *Survey (capture existing)...*): walk through an existing installation and type device, room, connection and note — Enter, next. Each entry becomes an unfinished device; the plan check asks for the missing details.
- **Label sheets and QR labels** for cables and devices.
- **Read-only web viewer** for people who do not run the app.

17. Working together: live session and cloud

Live session

- Real-time co-editing over WebRTC; no server holds your plan.
- Join by **invite link** or find sessions on the LAN.
- Set a **room password** — it encrypts the session end to end.

- Across networks the default relay `wss://relay.zumpelars.de` connects the peers; you can run your own relay and STUN/TURN servers ([self-hosted relay](#)) or use **local-only** mode.
- Undo takes back your own edits, not those of others.

Cloud copy with revisions

File → *Cloud & share link...* (requires a device library account, see chapter 18).

- Your file stays the master copy; the cloud keeps a copy with history.
- Every save is a revision — automatically 30 s after the last change, or with **Save to cloud now**. Any revision can be restored.
- Parallel changes from another device are merged, never overwritten.
- Credentials stay on the computer. **Delete from cloud** removes the project with all revisions.

18. Device library

The shared device catalogue of the planner suite: devices.zumpelars.de.

1. Create an account on the website, confirm your email and accept the guidelines.
2. Sign in under *Settings* → *Device library* (email or user name, password, two-factor code if enabled).
3. **Update from device library** fetches what changed since the last sync. The devices stay available offline.
4. Your own templates are uploaded automatically while you are signed in (*Upload my own devices automatically*), or with **Sync now**. *My devices* shows the state of each: not uploaded, waiting for moderation, live, blocked with reasons.

Submitting templates: *Library* → *+* → *Submit templates...* checks your templates first. Required: a datasheet link, and a connector type and label on every port. Power draw is optional. Templates that fail are listed with the reason and not sent.

19. Import and exchange

Source	What comes in
Rentman	projects, equipment, categories (selective)
NetBox	racks, devices, cable paths
GraphML	diagrams, with preview
MultiCam Planner (<code>.cameras.json</code>)	cameras with ports, lens, focal length, position and PTZ presets; importing again reconciles instead of duplicating
<code>.avplan</code>	shared exchange format of the planner suite
Venue (<code>venue-exchange</code>)	floor plan with scale, from MultiCam / Light Planner
Building statement (<code>.avfacility</code>)	outlets, cable routes, floors, house runs and control addresses from the Facility Planner
Green-GO	intercom export <code>.gg5</code> plus a vendor-neutral intercom file
Racks for the warehouse	<code>rack-belegung.json</code> for the Inventory Planner

API tokens are stored in the operating system's credential store (macOS Keychain, Windows Credential Manager, libsecret), never in the project file.

Inventory & stock (*Tools* → *Inventory* / *stock...*): warehouse stock with locations, serial numbers and quantities, checked against the plan's demand.

20. Claude (MCP)

Claude can answer questions about the open plan — devices, ports, signal paths, cables and the plan check.

On this computer

1. *Settings* → *MCP* → switch on. The server listens on `127.0.0.1` only and uses a pairing token from the keychain.
2. Copy the command shown there, for example:

```
claude mcp add --transport http cable-planner http://127.0.0.1:<port>/mcp --header "Authorization: Bearer <token>"
```

1. Reading is the default. **Writing** is a second switch: Claude can then connect and remove cables, set cable details and rename devices. Each call is one undo step and is listed under *What Claude changed*.

From `claude.ai`, the phone or another machine

1. Put the project into the cloud (*File* → *Cloud & share link...*).

2. In claude.ai: *Settings* → *Connectors* → *Add custom connector* with `https://devices.zumpelars.de/mcp` and sign in with your device library account.
3. Read-only, over your own cloud projects. Disconnect under *Account* → *Security* → *Connected apps* on `devices.zumpelars.de`.

Switching commands for ATEM or Videohub are never offered.

21. Web edition and tablet

- Open `https://larszu.github.io/cable-planner/` and add it to the home screen; it runs full-screen with its own icon.
- Touch: pinch zooms the plan, two fingers pan, a **long press** opens the context menu.
- Not available in the browser (no socket, listening port or keychain): ATEM, Videohub, NetBox, LAN sync, phone access, MCP server, show control, switching, Tally-Pi, Rentman export, update check. *Settings* → *Integrations* lists them with the reason.

22. Data, safety and troubleshooting

- **Projects**: local JSON files. The app writes them atomically with a backup copy.
- **App data** (library, recent projects, settings) lives in the app-data folder `Cable Planner`.
- **Secrets** (API tokens, stream credentials, device library sign-in) live in the operating system's keychain. The web edition keeps the device library sign-in in the browser's storage.
- **"No recovery copy"** in the status bar: save the project to a file.
- **No still image preview**: install ffmpeg, and check the device is in the local network.
- **Device library unreachable**: the app keeps working with the devices from the last sync.

Questions and bug reports: [GitHub Issues](#).

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